

Forecasting the Conditional Distribution of Realized Volatility of Oil Price Returns: The Role of Skewness over 1859 to 2023

Rangan Gupta^{*}, Qiang Ji^{**}, Christian Pierdzioch^{***} and Vasilios Plakandaras^{***}

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JEL Codes: C22, C53, Q02

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1. Introduction

Expected skewness is a measure of risk of the oil market as it captures the expected asymmetry of the distribution of future realizations of returns. Intuitively, variation in expected skewness is likely to originate from extreme directed changes in aggregate demand and supply (Yu et al., 2022; Salisu et al., 2022a), geopolitical acts and threats (Salisu et al., 2021), rare disaster events including pandemics (Bouri et al., 2020; Qin et al., 2020), and even financial market spillovers (Salisu et al., 2022b). Such changes, in turn, are likely to result in an impact on the volatility of the oil-price via a “leverage effect” in the oil market, as initially established by Geman and Shih (2009) and, more recently, by Asai et al. (2019, 2020). For instance, an increase in oil returns due to a supply shock would be associated with an economic slowdown, and this recessionary impact would likely reduce demand and trading activity in the oil market, leading to lower volatility in line with the leverage effect.

In sum, positive (negative) expected skewness in oil returns, on purely theoretical grounds, could be associated with decreasing or increasing (increasing or decreasing) subsequent oil price returns volatility. Hence, the sign of the link between expected skewness and subsequent

volatility can only be recovered empirically. In this paper, we predict both in- and out-of-sample the realized volatility (RV) of oil price returns based on expected skewness by utilizing a quantiles-based predictive regression model. With the quantile-regression model, unlike in the case of the popular Markov-switching and smooth transition threshold models, we do not need to specify *ad hoc* the number of regimes of RV . Furthermore, the quantile-regression model retains the simple structure of a linear predictive regression model for any given quantile of RV , but simultaneously, renders possible to add an element of non-linearity to our empirical research strategy in that the coefficients of the predictive model are allowed to vary across the different quantiles of the conditional distribution of RV . In light of the skewed distribution of RV , and statistical evidence of nonlinearity and structural breaks that we detect in its relationship with expected skewness, the derivation of our empirical results from a quantile-regression model, thus, is highly warranted.

In this regard, by covering the longest possible data available for constructing a measure of expected skewness over the historical period from 1859 to 2023, we capture various positive and negative oil shocks associated with, for example, the U.S. Civil War, the two World Wars, the West coast gas famine, the Great Depression, the Korean conflict, the Suez Crisis, the OPEC oil embargo, the Iranian revolution, the Iran-Iraq War, the Gulf War, the global financial crisis, the outbreak of the Coronavirus pandemic in 2020, and, of course, more recently the ongoing Russia-Ukraine War.¹ Moreover, upon studying the predictability of oil-price volatility due to oil risks associated with expected skewness, we avoid the issue of sample selection bias, and we provide a comprehensive answer to the question regarding the sign of the relationship between expected skewness and oil market volatility.

While in recent literature realized skewness, as derived from intraday data, has been utilized to predict daily RV for over the last couple of decades only (see for example, Gkillas et al., 2020, Demirer et al., 2022), we are the first to shed light on the importance of expected skewness in driving RV over the entire history of the WTI oil market spanning nearly 165 years. In this process, we add to the already voluminous literature on the predictability of oil market volatility based on a wide array of models, and macroeconomic, financial, behavioural, and climate patterns-related predictors (see, Bouri et al., 2021 for detailed reviews), by considering the role of expected skewness. The only related paper to ours is one by Salisu et al. (2022a), who report that negative tail risks, derived from a Conditional Autoregressive Value at Risk (CAViaR)

¹ The reader is referred to Hamilton (2013) for a detailed discussion of historical oil shocks from 1859 to 2009.

framework, can be utilized to the predict realized variance of oil returns over the monthly period of 1859:10 to 2020:10. Allowing skewness to capture both extreme negative and positive shocks, and utilizing a quantile-regression model to predict the entire conditional distribution of oil market volatility, is a technical improvement over the conditional mean-based negative shocks-only framework studied by Salisu et al. (2022a). At the same time, our findings should be of value to researchers, investors and policymakers, because our analysis nests Salisu et al. (2022a) in terms of a broader treatment of the shocks investigated, as well as the information on predictability involving the entire conditional distribution of RV .

The remainder of the paper is organized as follows: Section 2 describes the data and methodology, Section 3 discusses the forecasting results, and Section 4 concludes.

2. Data and Methodology

We use daily and monthly WTI crude oil prices obtained from Global Financial Data², over the period of 1859:11 to 2023:04. Both frequencies of the oil price data are converted into log returns in percentage, i.e., the first-difference of the natural logarithm of the price multiplied by 100. Then RV is computed as the square root of the sum of daily squared returns over a month, as per availability of daily data. Note that oil price data is available at monthly frequency only over the period 1920-1976 and, hence, we measure RV as the square root of monthly squared log-returns over this part of the overall sample.

We consider the specific quantile regression model of the following form:

$$RV_{t+1} = \beta_0^{(\tau)} + \beta_1^{(\tau)}RV_t + \beta_2^{(\tau)}RV_{t-1} + \beta_3^{(\tau)}RV_{t-2} + \beta_4^{(\tau)}SKEWNESS_t + \varepsilon_{t+1} \quad (1)$$

where RV_t is the realized volatility at time t ; $SKEWNESS_t$ is the expected skewness at time t (which we discuss below in detail); $\tau \in (0,1)$ and ε_{t+h} are assumed independent derived from

an error distribution $g_\tau(\varepsilon)$ with the τ -th quantile equal to 0, i.e., $\int_{-\infty}^0 g_\tau(\varepsilon)d\varepsilon = \tau$. The three lags of RV are selected based on the Schwarz Information Criterion (SIC), and capture the persistence of the process, with the intercept and the regression coefficients depending upon τ .

The estimators of the parameters of the quantile regression model in Equation (1) minimize $\sum_{t=0}^{T-h} \rho_\tau(RV_{t+1} - \beta_0^{(\tau)} + \beta_1^{(\tau)}RV_t + \beta_2^{(\tau)}RV_{t-1} + \beta_3^{(\tau)}RV_{t-2} + \beta_4^{(\tau)}SKEWNESS_t)$, where $\rho_\tau(u) = u(\tau - I(u < 0)) = \frac{1}{2}[|u| + (2\tau - 1)u]$. The forecast of the τ -th quantile of the distribution RV at time $t + 1$ is obtained as: $\widehat{RV}_{t+1}(\tau) = \widehat{\beta}_0^{(\tau)} + \widehat{\beta}_1^{(\tau)}RV_t + \widehat{\beta}_2^{(\tau)}RV_{t-1} +$

² <https://globalfinancialdata.com/>.

$\widehat{\beta}_3^{(\tau)}RV_{t-2} + \widehat{\beta}_4^{(\tau)}SKEWNESS_t$. Our benchmark model is nested in equation (1) and obtained by setting $\widehat{\beta}_4^{(\tau)} = 0$.

In order to get a time-varying measure of expected skewness of oil returns (r), we estimate the following autoregressive quantile regression of order 1:

$$Q^\tau(r_t) = \gamma_0^\tau + \gamma_1^\tau(r_{t-1}) \quad (2)$$

Using the estimated model parameters from this quantile regression, we compute the one-step-ahead expected, or predicted, Kelley skewness (Kelley, 1947) as follows:

$$\mathbb{E}_{t-1}[SKEWNESS(r_t)] = \frac{\mathbb{E}_{t-1}[Q_t^{0.9}] + \mathbb{E}_{t-1}[Q_t^{0.1}] - 2\mathbb{E}_{t-1}[Q_t^{0.5}]}{\mathbb{E}_{t-1}[Q_t^{0.9}] - \mathbb{E}_{t-1}[Q_t^{0.1}]} \quad (3)$$

In equation (1), we simplify the notation by setting $\mathbb{E}_{t-1}[SKEWNESS(r_t)]$ to $SKEWNESS_t$ henceforth.

Figure A1 (Appendix) plots RV_t and $SKEWNESS_t$, with Table A1 presenting the corresponding summary statistics. The variables are non-normal distributed based on the rejection of the null hypothesis of normality under the Jarque-Bera test at the highest level of significance, with the heavy right tail of RV providing a preliminary motivation to look at a quantile predictive regression model.

3. Empirical Results

For the sake of completeness and comparability, we also estimate the conditional mean version of equation (1) using ordinary least squares (OLS) with heteroscedasticity and autocorrelation adjusted (HAC) standard errors. Findings (not reported in detail) show that $\widehat{\beta}_4$ is equal to -12.2154, with a p -value of 0.4095, i.e., the predictive effect of $SKEWNESS$ is found to be negative but insignificant.³

In order to motivate the quantile regression approach, we now turn to statistical evidence of possible misspecification in the classical mean regression model, which may account for the insignificant predictive impact of expected skewness. In this regard, we first apply the Brock et al. (1996, BDS) test of nonlinearity on the residuals of the conditional mean model outlined

³ Following Ghysels et al. (2018), we also obtained the measure of $SKEWNESS$ based on a quantile-Mixed Data Sampling (MIDAS) model. However, the estimate of β_4 ($=-2.6273$), continued to be insignificant (with a p -value of 0.3086), with further details of this exercise are available upon request from the authors. This observation tends to suggest that the underlying way of computing the expected skewness is not necessarily the source of insignificant predictability, but, as we show below with the quantile predictive model, it is more due to misspecifications with the conditional mean framework.

above. As can be observed from Table 1, the *i.i.d.* null hypothesis is rejected for all dimensions (m) at the highest level of significance, thus providing strong evidence of nonlinearity in the relationship between $SKEWNESS_t$ and RV_{t+1} . In addition, we also conducted the Bai and Perron (2003) test of multiple structural breaks on the conditional-mean estimated regression. The $UDmax$ and $WDmax$ tests under the specification of 1 to M globally determined breaks revealed five breaks at: 1884:11, 1909:04, 1933:10, 1966:11, and 1991:04. Existence of nonlinearity and regime-changes confirm that the linear conditional mean-based predictive regression is misspecified, and the (insignificant) predictive impact of expected skewness on the RV cannot be considered reliable. This, in turn, leads us to consider the quantile regression model for further analyses.

[INSERT TABLE 1 HERE]

In Figure 1, we present $\widehat{\beta}_4^{(\tau)}$, at $\tau = 0.05, 0.10, \dots, 0.90, 0.95$, i.e., the estimated quantile-specific response of $SKEWNESS_t$ and RV_{t+1} . As it is obvious, the effect is positive until quantile 0.40, and then turns negative. We observe a significant (at 1% level) positive effect over the quantile range of 0.35-0.40, while the predictive impact is significantly negative at the highest level over $\tau = 0.50$ to 0.95.⁴ In other words, the sign and statistical significance of predictability of RV due to $SKEWNESS$ is quantile-specific. Economically speaking, when risk is initially low in the oil market, as captured by the lower quantiles, a positive expected skewness causing higher returns seems to be driving more trading activity and, hence, volatility, as in Demirer et al. (2020). However, beyond the median, higher returns associated with a higher skewness, tend to be associated with lower uncertainty, and reduce the risk in the oil market via a well-established leverage effect in WTI oil prices, reported, among others, in Agnolucci (2009), Larsson and Nossman (2011). Put differently, we observe that the leverage effect causes a negative link between expected skewness and oil market volatility only beyond the conditional median of the latter. Using the longest available data sample, we are thus able to add another layer to the volatility-skewness nexus of oil returns in the sense that not only do we find positive

⁴ Figure A2 (Appendix) plots the predictive impact of expected kurtosis on RV . Note that, after estimating Equation (2), $\mathbb{E}_{t-1}[KURTOSIS(r_t)]$ is given by: $\frac{\mathbb{E}_{t-1}[Q_t^{0.99}] + \mathbb{E}_{t-1}[Q_t^{0.01}]}{\mathbb{E}_{t-1}[Q_t^{0.75}] - \mathbb{E}_{t-1}[Q_t^{0.25}]}$. The nature of predictability for RV_t due to $KURTOSIS_{t-1}$ is qualitatively similar to that of $SKEWNESS_{t-1}$, though in this case only statistically significant negative effect at the 1% level is detected over the quantile range of 0.45 to 0.95.

and negative impacts of the latter on the former, but these signs are contingent on the states, as defined by the quantiles, of the latter.⁵

[INSERT FIGURE 1 HERE]

Since in-sample predictability does not guarantee out-of-sample forecasting gains, we conduct a one-step-ahead ($h = 1$) forecasting exercise that starts in 1884:11 and corresponds to the first identified structural break date. The quantile regression is recursively estimated over the out-of-sample period of 1884:11-2023:04, with an in-sample of 1859:11-1884:10, for each of the 19 considered quantiles. In line with the extant literature of using (Root) Mean Square Errors ((R)MSE) as the forecast performance statistic involving quantile regressions in forecasting movements in financial and macroeconomic variables (Gupta et al., 2017; Korobilis, 2017), in Table 2, we present the *MSE-F* test statistic of McCracken (2007). As can be seen from the table, forecasting gains due to expected skewness for *RV* are primarily statistically significant, consistently at the 1% level, over the range $\tau = 0.60$ to 0.95 (with the exceptions of the quantiles 0.70 and 0.85), which is in line with the strongly significant leverage effect observed in Figure 1, associated with in-sample predictability.

Given that, the ultimate test of any predictive model (in terms of econometric methodologies and predictors used) is in its forecasting performance (Campbell, 2008), in Table 2, we present the *MSE-F* statistics for the longer forecasting horizons of 3-, 6- and 12-month-ahead over the out-of-sample period of 1884:11 to 2023:04. The quantile-regression model continues to be the preferred model in light of strong evidence of nonlinearity at the longer forecast horizons, as reported in Table 1. Furthermore, for $h = 3$ -, 6 -, and 12 -month-ahead, five break dates are detected for each case,⁶ and, hence, warrant the usage of the quantiles-based approach. The results are in line with the one-month-ahead i.e., $h = 1$ findings, especially for $h = 3$, with the effect being slightly weaker under $h = 6$, but quite stronger under $h = 12$.⁷ From these results, we can, in general, conclude that expected skewness forecasts the extreme ends and around the

⁵ In this regard, note that, Kristoufek (2014) had indicated that the existence of the leverage effect is time-scale-dependent.

⁶ The breaks were at: 1884:11, 1909:04, 1933:10, 1966:11, 1991:04; 1885:04, 1909:09, 1934:03, 1974:05, 1998:10, and; 1885:08, 1909:12, 1934:08, 1961:12, 1986:04 for $h = 3$ -, 6 -, and 12 -month-ahead, respectively.

⁷ Note that, for $h > 1$, RV_{t+h} is the average value of RV over t to $t+h$, and then the same static-forecasting approach over the out-of-sample period.

median of the conditional distribution of oil market volatility, especially in the longer-run.⁸ In the process, just like intraday data-based results spanning the last two decades or so, skewness is shown to contain information for producing accurate forecasts of RV on oil price returns, but only in a quantile-specific manner.⁹

[INSERT TABLE 2 HERE]

4. Conclusions

We analyse the predictive value of the expected skewness of oil returns for the realized volatility of oil price returns over the historical monthly period of 1859:11 to 2023:04. Our in-sample analysis shows that, while predictability cannot be detected by the misspecified conditional mean-based predictive model, a causal impact does indeed exist in a quantile-regression model, that is able to accommodate for such misspecifications. Specifically, we observed that the predictive effect of skewness can be both positive and negative, with the sign contingent on the states (quantiles) of realized volatility. Based on an out-of-sample forecasting analysis, we also found statistically significant forecasting gains not only at the short one-step-ahead horizon, but also at medium- to long-run forecasting horizons of three-, six-, and twelve-month-ahead.

Our findings can be used by policymakers to obtain information on the future path of oil returns volatility due to associated expected skewness at various forecasting horizons. This knowledge, in turn, could be useful to predict economic recessions gathering steam, given that oil-price volatility is known to negatively impact real economic activity, by designing and implementing appropriate response of macroeconomic policies. Moreover, the forecastability of oil returns volatility due to expected skewness should be of vital importance to oil investors, especially over the investment horizon of one year. Finally, academically, we add to the debate of whether

⁸ Interestingly, the $MSE-F$ statistics for the conditional mean-based model was also found to be statistically significant at the 1% level for $h = 1, 3$, and 6 , but not 12 , with the stats being respectively, 20.6915 , 20.6667 , 5.9406 , and -4.7934 . The fact that the misspecified model produced forecasting gains for 3 of the 4 horizons, could be due to the fact that the recursive estimation of the conditional mean model over the out-of-sample period starts on the first break date, and hence, is able to accommodate, at least partially, for nonlinearity and regime changes, unlike in the case of the full-sample predictability test.

⁹ We must point out that, the quantile-MIDAS-based estimate of $SKEWNESS$ as outlined in Ghysels et al. (2018) performed relatively weakly in the sense that it produced significant forecasting gains for RV in 23 cases compared to the 33 derived under the standard $AR(1)$ quantile regression approach in Equation (2), for the four forecasting horizons considered. Complete details of these results are available upon request from the authors.

oil price returns skewness impacts the volatility of the market negatively or positively, by showing that both these effects are possible.

As an extension of this study, it would be interesting to investigate the role of own- and/or cross-market(s) expected skewness in predicting historical volatility of other important asset markets, contingent on the availability of data.

References

- Agnolucci, P. (2009). Volatility in crude oil futures: A comparison of the predictive ability of GARCH and implied volatility models. *Energy Economics*, 31(2), 316-321.
- Asai, M. Gupta, R., and McAleer, M. (2019). The Impact of Jumps and Leverage in Forecasting the Co-Volatility of Oil and Gold Futures. *Energies*, 12, 3379.
- Asai, M. Gupta, R., and McAleer, M. (2020). Forecasting Volatility and co-volatility of crude oil and gold futures: Effects of leverage, jumps, spillovers, and geopolitical risks. *International Journal of Forecasting*, 36(3), 933-948.
- Bai, J., and Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18, 1-22.
- Bouri, E., Gkillas, K., Gupta, R., and Pierdzioch, C. (2020). Infectious diseases, market uncertainty and realized volatility of oil. *Energies*, 13(16), 4090.
- Bouri, E., Gupta, R., Pierdzioch, C., and Salisu, A.A. (2021). El Niño and forecastability of oil-price realized volatility. *Theoretical and Applied Climatology*, 144(22), 1173-1180.
- Brock, W.A., Scheinkman, J.A., Dechert, W.D., and LeBaron, B. (1996). A test for independence based on the correlation dimension. *Econometric Reviews*, 15(3), 197-235.
- Campbell, J.Y. (2008). Viewpoint: Estimating the equity premium. *Canadian Journal of Economics*, 41(1), 1-21.
- Demirer, R., Gupta, R., Pierdzioch, C., and Shahzad, S.J.H. (2020). The predictive power of oil price shocks on realized volatility of oil: A note. *Resources Policy*, 69(C), 101856.

Geman, H., and Shih, Y.F. (2009). Modeling Commodity Prices under the CEV Model. *Journal of Alternative Investments*, 11(3), 65-84.

Gkillas, K., Gupta, R., and Pierdzioch, C. (2020). Forecasting realized oil-price volatility: The role of financial stress and asymmetric loss. *Journal of International Money and Finance*, 104(C), 102137.

Gupta, R., Majumdar, A., and Wohar, M.E. (2017). The Role of Current Account Balance in Forecasting the US Equity Premium: Evidence From a Quantile Predictive

Korobilis, D. (2017). Quantile regression forecasts of inflation under model uncertainty. *International Journal of Forecasting*, 33(C), 11-20.

McCracken, M.W. (2007). Asymptotics for out of sample tests of Granger causality. *Journal of Econometrics*, 140, 719-752.

Larsson, K., and Nossman, M. (2011). Jumps and stochastic volatility in oil prices: Time series evidence. *Energy Economics* 33(3), 504-514.

Qin, M., Qiu, L-H., Tao, R., Umar, M., Su, C-W., and Jiao, W. (2020). The inevitable role of El Niño: a fresh insight into the oil market. *Economic Research-Ekonomska Istraživanja*, 33(1), 1943-1962.

Salisu, A.A., Gupta, R., and Ji, Q. (2022b). Forecasting oil prices over 150 years: The role of tail risks. *Resources Policy*, 75(C), 102508.

Salisu, A.A., Pierdzioch, C., and Gupta, R. (2021). Geopolitical risk and forecastability of tail risk in the oil market: Evidence from over a century of monthly data. *Energy*, 235, 121333.

Salisu, A.A., Pierdzioch, C., and Gupta, R. (2022a). Oil Tail Risks and the Forecastability of the Realized Variance of Oil-Price: Evidence from Over 150 Years of Data. *Finance Research Letters*, 46(B), 102378.

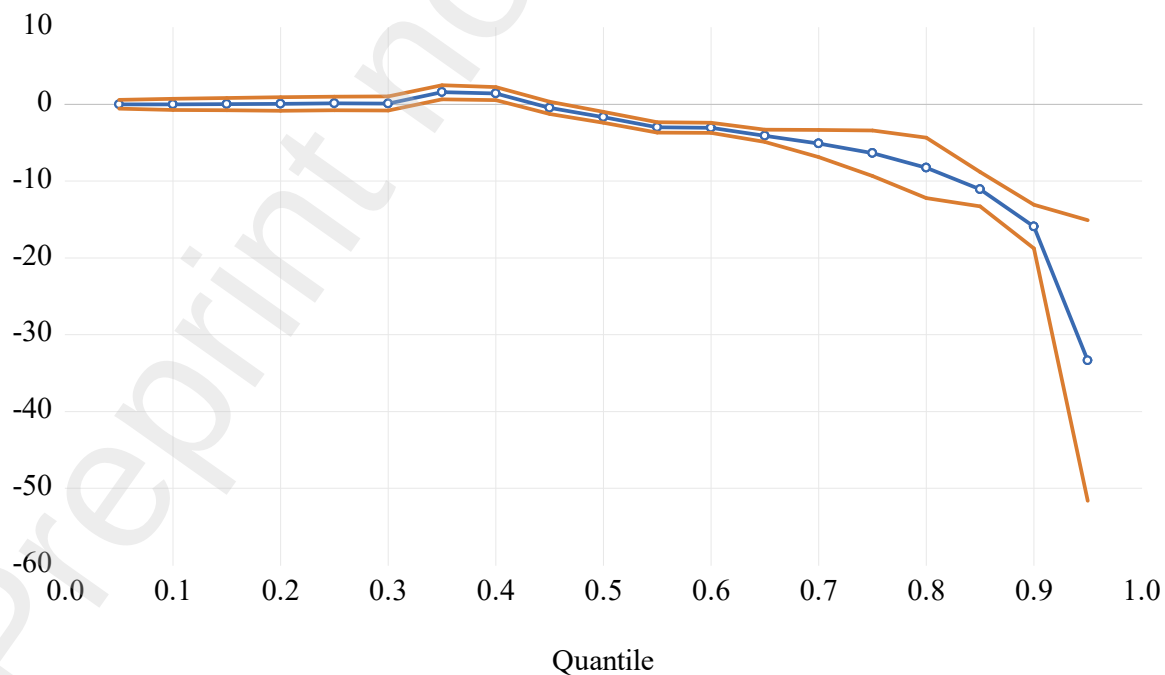
Yu, X., Wang, X., Wang, Y., and Li, Y. (2022). The effects of skewness on hedging decisions: an application of the skew-normal distribution in WTI and Brent futures, *Economic Research-Ekonomska Istraživanja*, 35(1), 3099-3118.

Table 1. BDS Test of Nonlinearity

m	$h = 1$	$h = 3$	$h = 6$	$h = 12$
2	23.5949***	23.7299***	33.3961***	45.7055***
3	26.6924***	26.8075***	36.5895***	50.0014***
4	28.4299***	28.6670***	38.9465***	53.7848***
5	30.8409***	31.1933***	43.2056***	59.1051***
6	33.9820***	34.6113***	48.4391***	65.8200***

Note: m stands for the number of (embedded) dimension which embed the time series into m -dimensional vectors, by taking each m successive points in the series; Entries correspond to the z -statistic associated with the BDS test applied to the residuals recovered from the conditional mean model: $RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-1} + \beta_3 RV_{t-2} + \beta_4 SKEWNESS_t + \varepsilon_{t+h}$; h is the forecast horizon; *** indicates rejection of the null hypothesis of *i.i.d.* residuals involving the BDS test at 1% level of significance.

Figure 1. In-sample Predictive Impact of Expected Skewness (*SKEWNESS*) on Realized Volatility (*RV*)



Note: The figure plots the estimated parameter $\hat{\beta}_4^{(\tau)}$ (in blue) for the equation: $RV_{t+1} = \beta_0^{(\tau)} + \beta_1^{(\tau)}RV_t + \beta_2^{(\tau)}RV_{t-1} + \beta_3^{(\tau)}RV_{t-2} + \beta_4^{(\tau)}SKEWNESS_t + \varepsilon_{t+1}$ at the quantiles $\tau = 0.05, 0.10, \dots, 0.90, 0.95$, with lower and upper 95% confidence bands (in orange).

Table 2. Out-of-sample Forecasting Results of Realized Volatility (RV) from Expected Skewness (SKEWNESS)

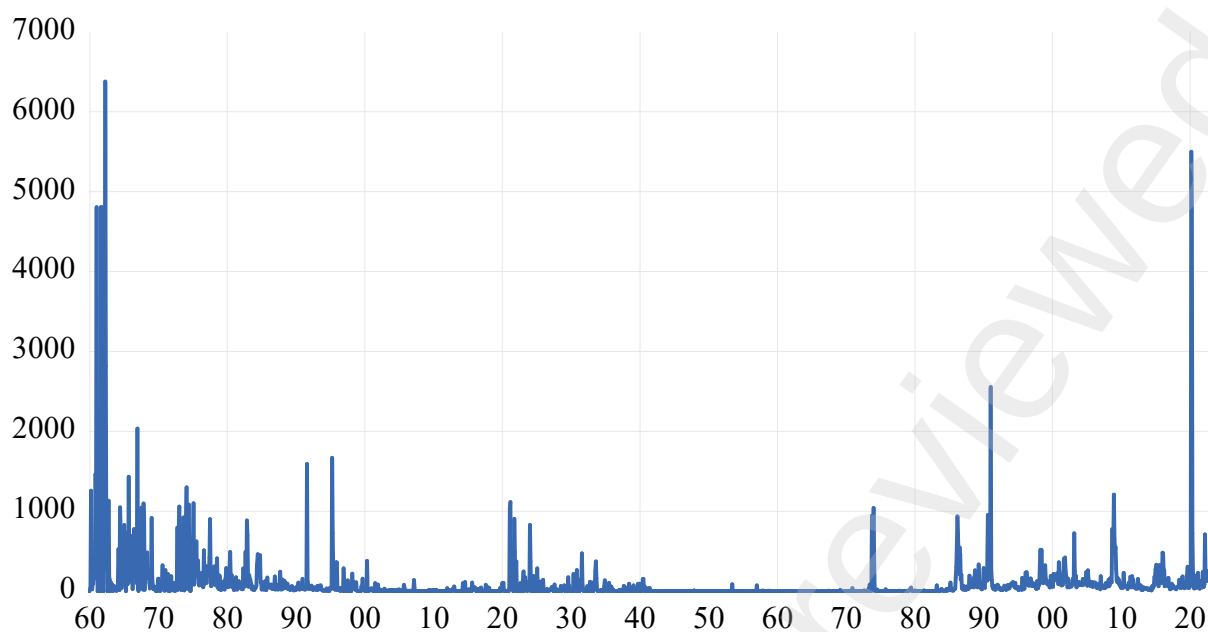
Quantile Regression (τ)	$h = 1$	$h = 6$	$h = 9$	$h = 12$
0.05	5.9417***	5.9345***	4.0521***	0.3690
0.10	4.0595***	4.0546***	33.4010***	1.7292**
0.15	1.9969**	1.9945**	-22.9396	6.0962***
0.20	-25.7436	-25.7127	-7.9035	-18.5431
0.25	0.3743	0.3738	-42.9027	-52.7130
0.30	-4.4436	-4.4383	-34.5828	-14.2317
0.35	-23.1330	-23.1052	-53.6389	-20.3499
0.40	-5.8396	-5.8326	-25.9456	7.5521***
0.45	-1.8121	-1.8099	-18.4494	109.4094***
0.50	-0.6746	-0.6738	-10.1691	25.0393***
0.55	-0.5909	-0.5902	16.6940***	20.2168***
0.60	15.2664***	15.2480***	-9.6048	-73.3776
0.65	89.4419***	89.3345***	-77.1870	112.2568***
0.70	-3.1969	-3.1931	-9.9420	97.4393***
0.75	120.6791***	120.5343***	-276.7012	-7.7630
0.80	120.6431***	120.4983***	-10.3778	16.5276***
0.85	-9.1227	-9.1117	-67.0148	7.8648***
0.90	14.8585***	14.8406***	13.9713***	-101.2330
0.95	47.4214***	47.3644***	-113.5328	12.9182***

Note: The entries correspond to the *MSE-F* test statistic of McCracken (2007), which tests whether the MSE of the unrestricted model ($RV_{t+h} = \beta_0^{(\tau)} + \beta_1^{(\tau)}RV_t + \beta_2^{(\tau)}RV_{t-1} + \beta_3^{(\tau)}RV_{t-2} + \beta_4^{(\tau)}SKEWNESS_t + \varepsilon_{t+h}$) is statistically lower than that of the restricted model ($RV_{t+h} = \beta_0^{(\tau)} + \beta_1^{(\tau)}RV_t + \beta_2^{(\tau)}RV_{t-1} + \beta_3^{(\tau)}RV_{t-2} + \varepsilon_{t+h}$) for a specific quantile τ and forecast horizon h ; *** and ** indicates 1% and 5% levels of significance respectively of the *MSE-F* statistic with critical values of 3.9510 and 1.5180.

APPENDIX:

Figure A1. Data Plot

(a). Realized Volatility (RV)



(b). Expected Skewness (*SKEWNESS*)

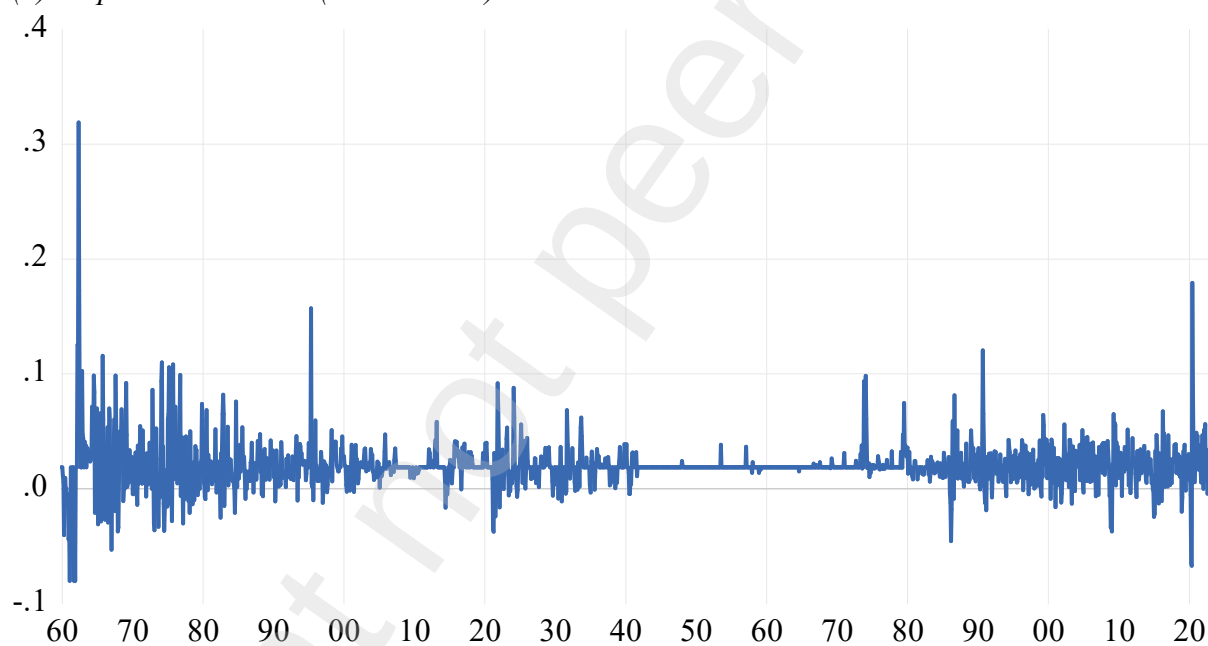


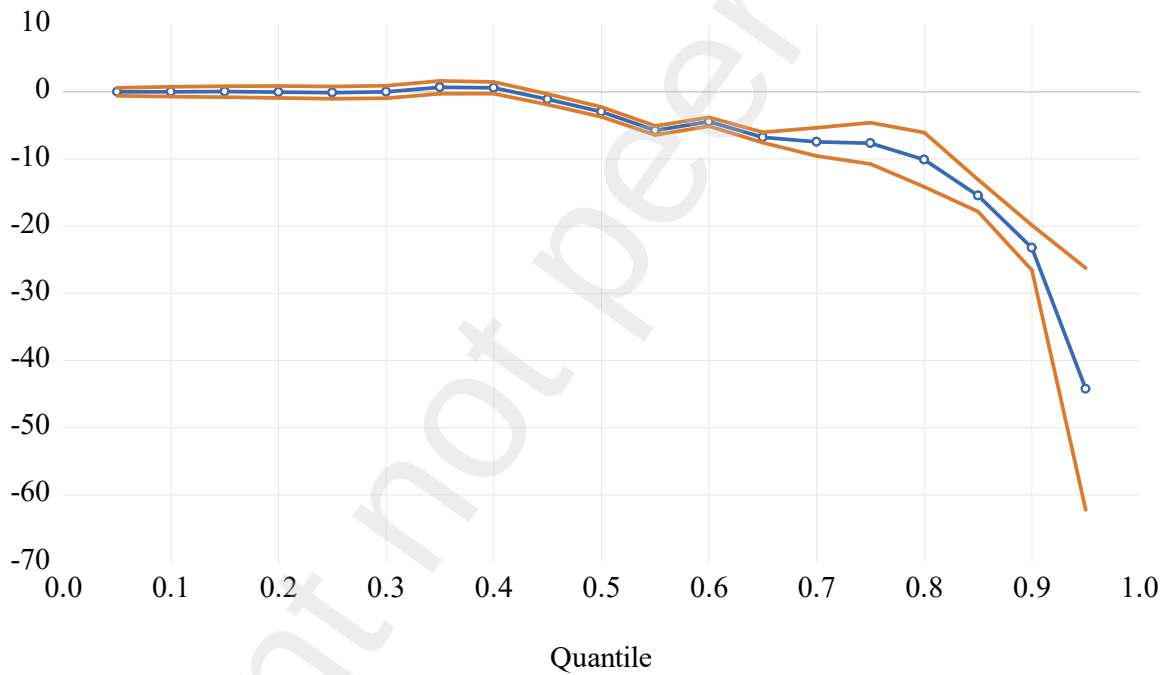
Table A1. Summary Statistics

STATISTIC	<i>RV</i>	<i>SKEWNESS</i>
Mean	101.1786	0.0197
Median	18.8682	0.0186

Maximum	6376.1450	0.3189
Minimum	0.0000	-0.0803
Std. Dev.	351.3405	0.0198
Skewness	10.7726	2.7898
Kurtosis	151.1006	38.7215
Jarque-Bera	1831036.0000	106860.1000
<i>p</i> -value	0.0000	0.0000
Obs.	1962	1962

Note: *RV* and *SKEWNESS* denotes realized volatility and skewness of oil returns respectively; Std. Dev. symbolizes the Standard Deviation; *p*-value corresponds to the test of normality based on the Jarque-Bera test.

Figure A2. In-sample Predictive Impact of Expected Kurtosis (*KURTOSIS*) on Realized Volatility (*RV*)



Note: The figure plots the estimated parameter $\hat{\beta}_4^{(\tau)}$ (in blue) for the equation: $RV_{t+1} = \beta_0^{(\tau)} + \beta_1^{(\tau)}RV_t + \beta_2^{(\tau)}RV_{t-1} + \beta_3^{(\tau)}RV_{t-2} + \beta_4^{(\tau)}KURTOSIS_t + \varepsilon_{t+1}$ at the quantiles $\tau = 0.05, 0.10, \dots, 0.90, 0.95$, with lower and upper 95% confidence bands (in orange).